

AWS Production Architecture Lab

Custom VPC + ALB + Target Group + Launch Template + Auto Scaling Group

1. Architecture Overview

Internet → ALB (Public Subnet)

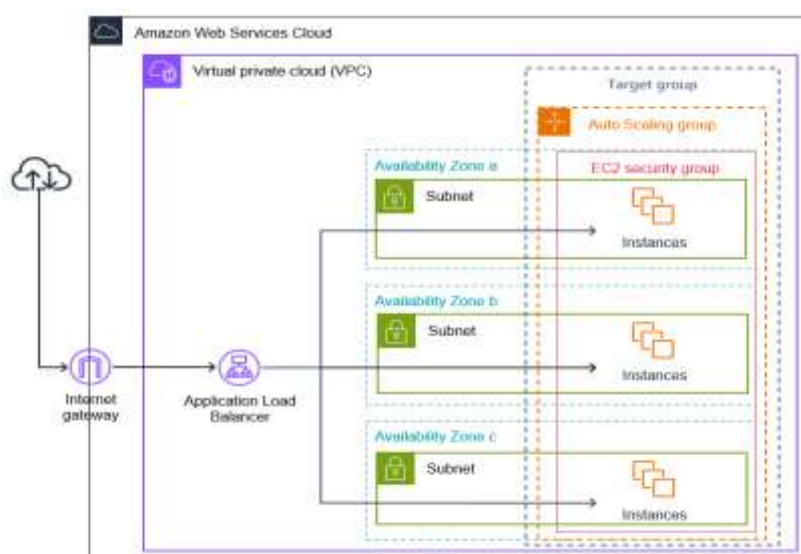
→ Target Group

→ Auto Scaling Group (Private Subnets)

→ EC2 Instances

Key Design Decisions

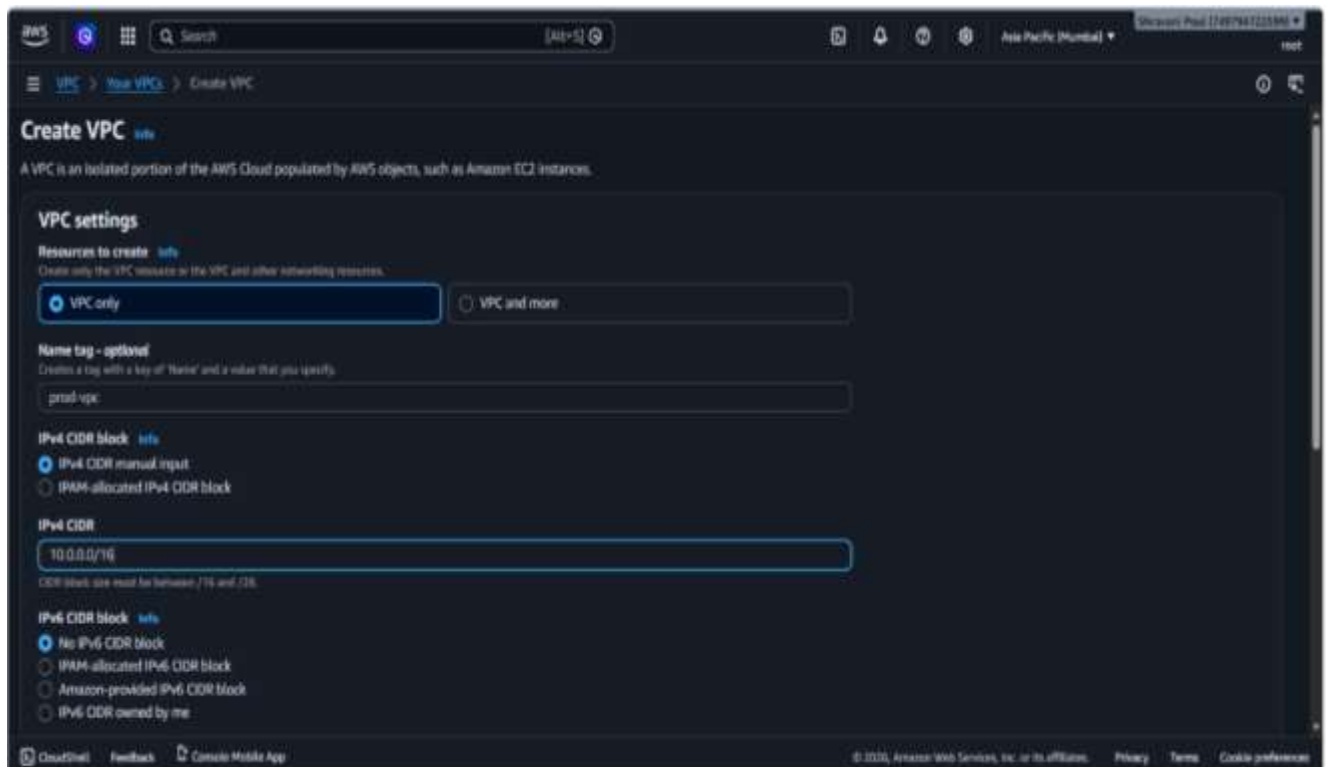
- ALB in **public subnet**
- EC2 in **private subnet**
- Multi-AZ deployment
- Auto Scaling enabled
- Health checks enabled



2. Step 1: Create Custom VPC

VPC Configuration

- Name: prod-vpc
- CIDR: 10.0.0.0/16

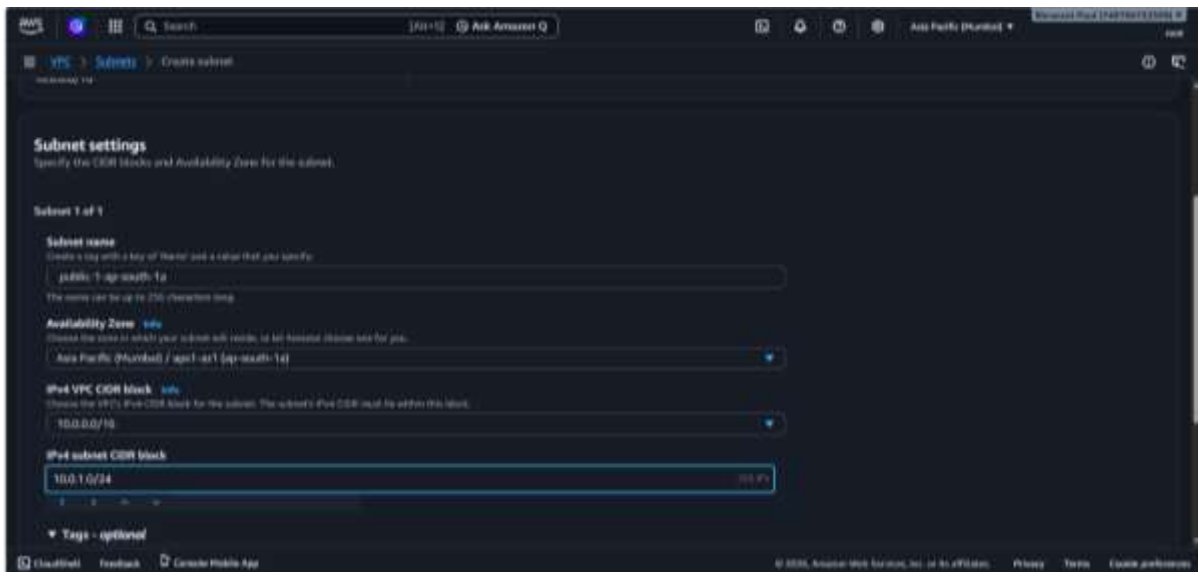


Create Subnets (VERY IMPORTANT)

We create 6 subnets across 3 AZs

Public Subnets:

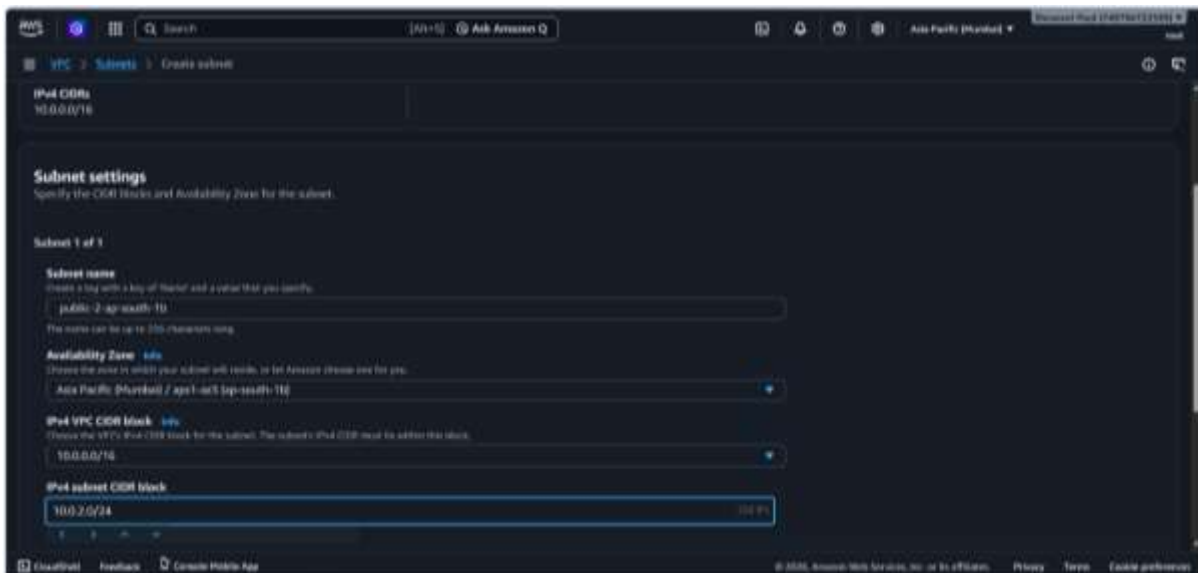
1. Name: public-1-ap-south-1a
subnet_Cidr: 10.0.1.0/24
Az: ap-south-1a



2. Name: public-2-ap-south-1b

subnet_Cidr: 10.0.2.0/24

Az: ap-south-1b



3. Name: public-3-ap-south-1c

subnet_Cidr: 10.0.3.0/24

Az: ap-south-1c

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of Name and a value that you specify.
public-3-ap-south-1c
The name can be up to 255 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24

Tags - optional

Key	Value - optional
-----	------------------

Private Subnets:

1. Name: private-1-ap-south-1a
subnet_Cidr: 10.0.11.0/24
Az: ap-south-1a

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of Name and a value that you specify.
private-1-ap-south-1a
The name can be up to 255 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.11.0/24

Tags - optional

Key	Value - optional
Name	private-1-ap-south-1a

2. Name: private-2-ap-south-1b

subnet_Cidr: 10.0.12.0/24

Az: ap-south-1b

The screenshot shows the 'Create subnet' page in the AWS Management Console. The 'Subnet settings' section is active, showing the following configuration for 'Subnet 1 of 1':

- Subnet name:** private-2-ap-south-1b
- Availability Zone:** Asia Pacific (Mumbai) / ap-south-1b
- IPv4 VPC CIDR block:** 10.0.0.0/16
- IPv4 subnet CIDR block:** 10.0.12.0/24
- Tags - optional:** A table with columns 'Key' and 'Value - optional'. The 'Key' column contains 'Name'.

The bottom of the page shows the 'Create' button and the footer with '© 2016, Amazon Web Services, Inc. or its affiliates'.

3. Name: private-3-ap-south-1c

subnet_Cidr: 10.0.13.0/24

Az: ap-south-1c

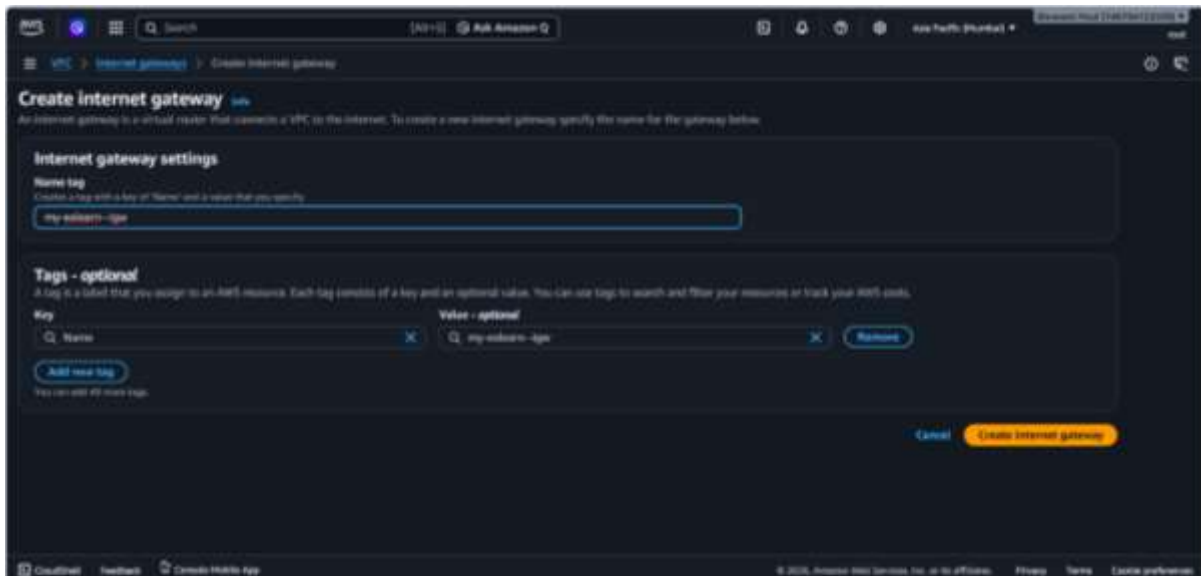
The screenshot shows the 'Create subnet' page in the AWS Management Console. The 'Subnet settings' section is active, showing the following configuration for 'Subnet 1 of 1':

- Subnet name:** private-3-ap-south-1c
- Availability Zone:** Asia Pacific (Mumbai) / ap-south-1c
- IPv4 VPC CIDR block:** 10.0.0.0/16
- IPv4 subnet CIDR block:** 10.0.13.0/24
- Tags - optional:** A table with columns 'Key' and 'Value - optional'. The 'Key' column contains 'Name'.

The bottom of the page shows the 'Create' button and the footer with '© 2016, Amazon Web Services, Inc. or its affiliates'.

3. Step 2: Create Internet Gateway (IGW)

- Create IGW
- Attach to VPC

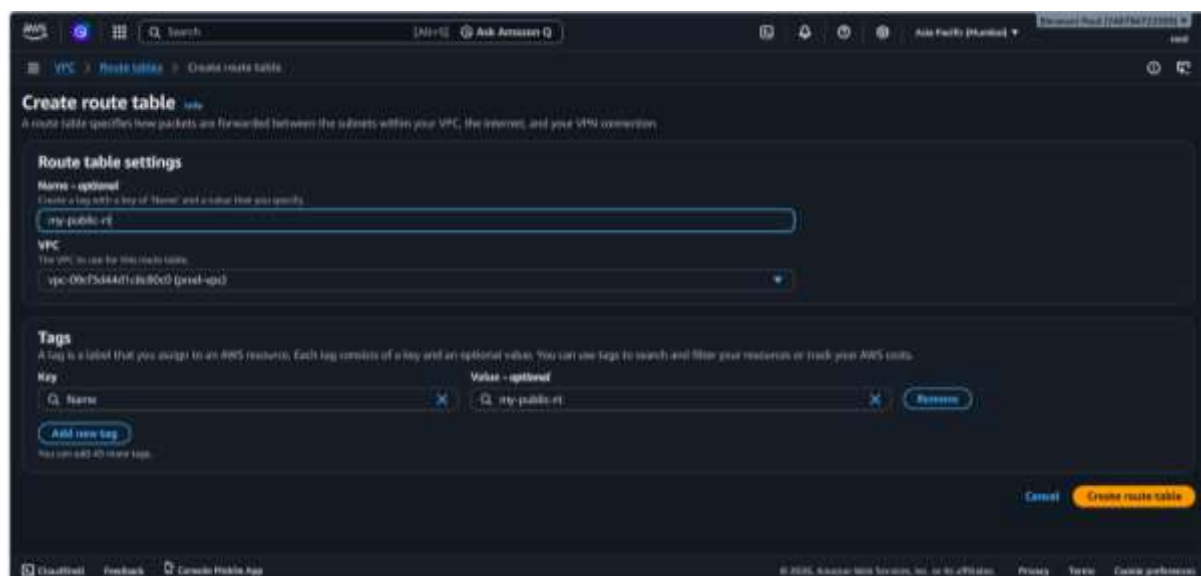


4. Step 3: Route Table for Public Subnets

Create route table

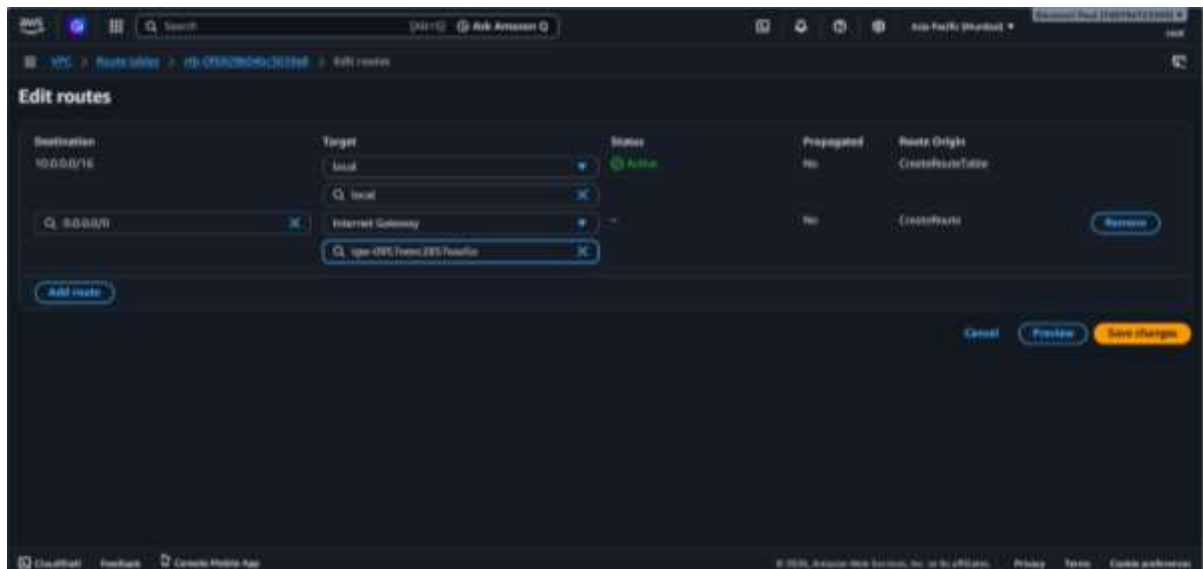
Name: my-public-rt

Vpc: choose my-prod-vpc



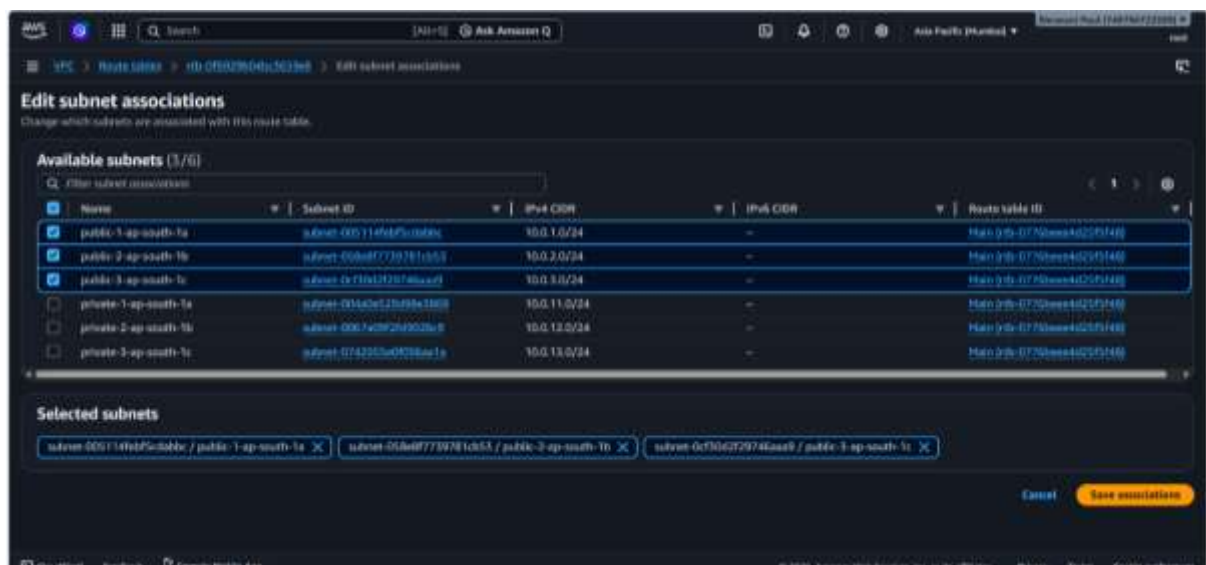
Add Route:

0.0.0.0/0 → Internet Gateway

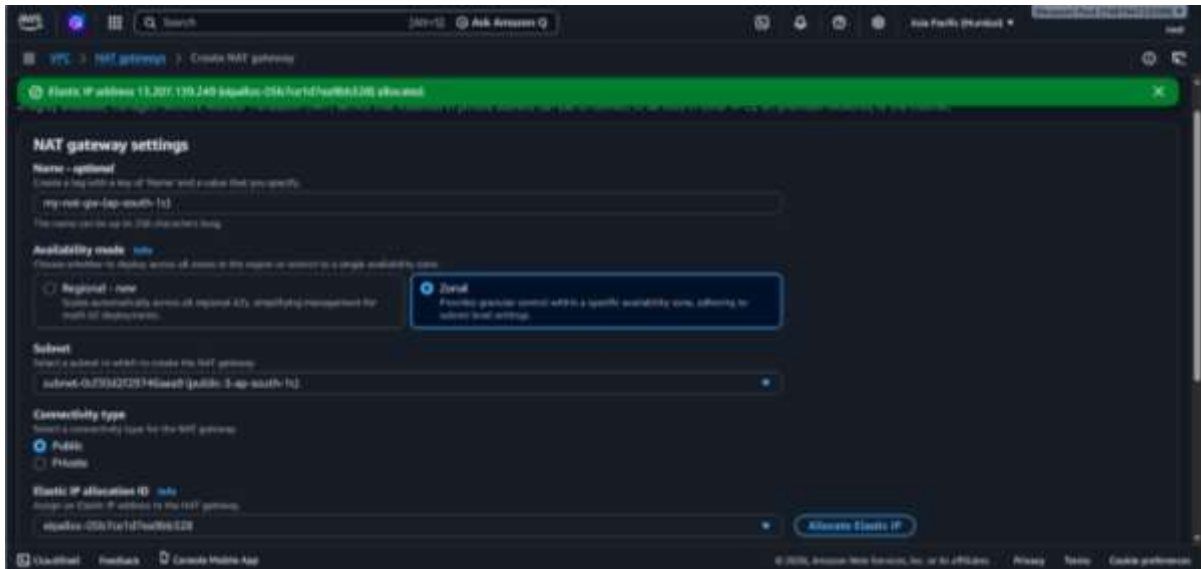


Associate:

- public-1
- public-2
- public-3



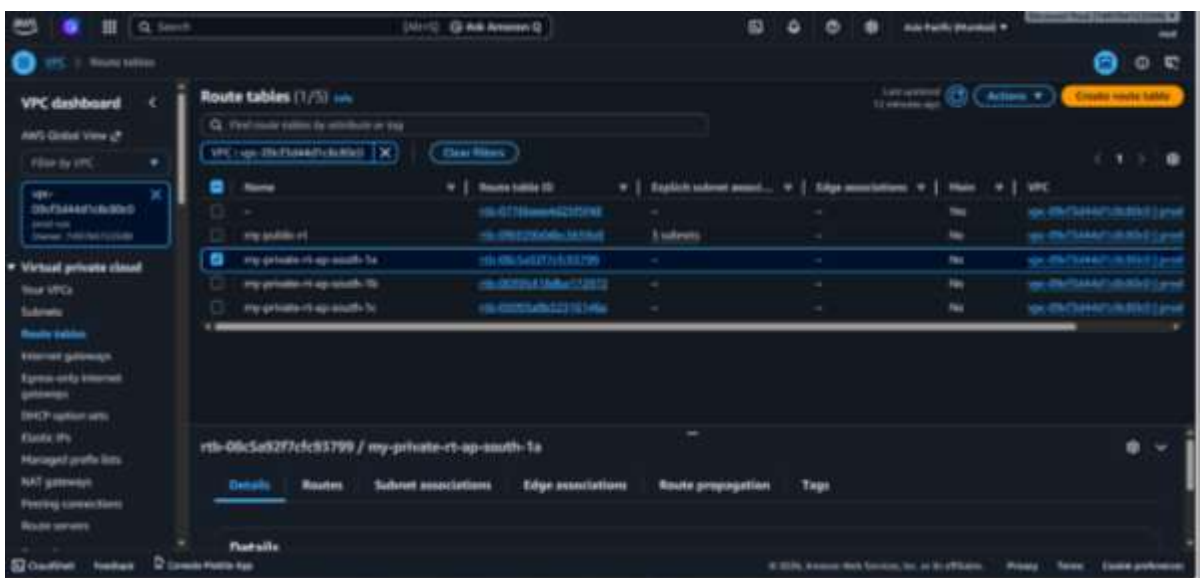
5. Step 4: Create NAT Gateways (CRITICAL)

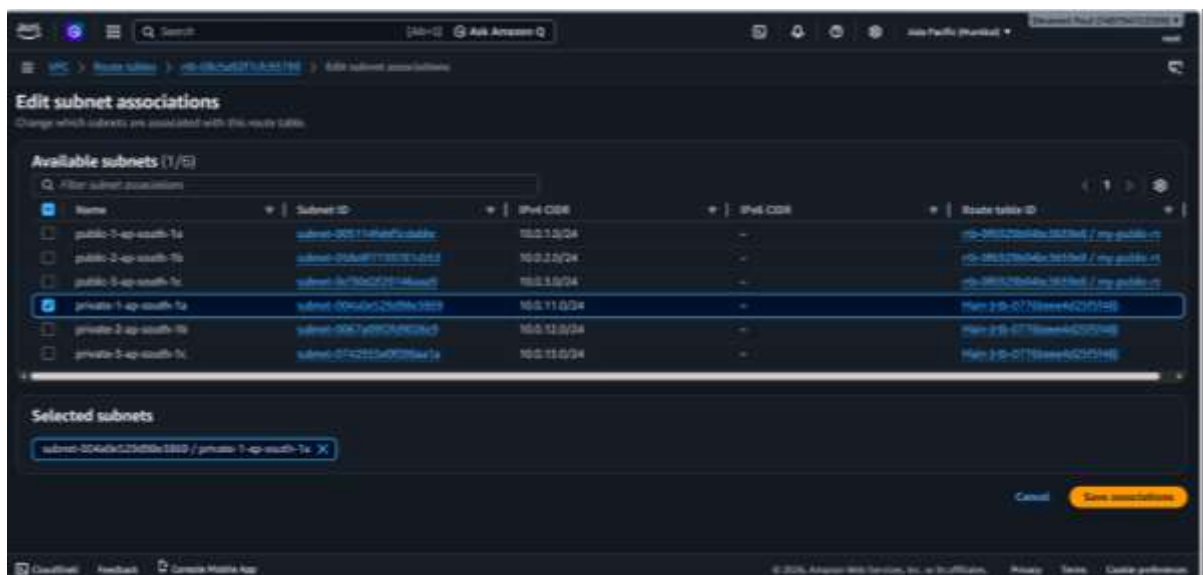
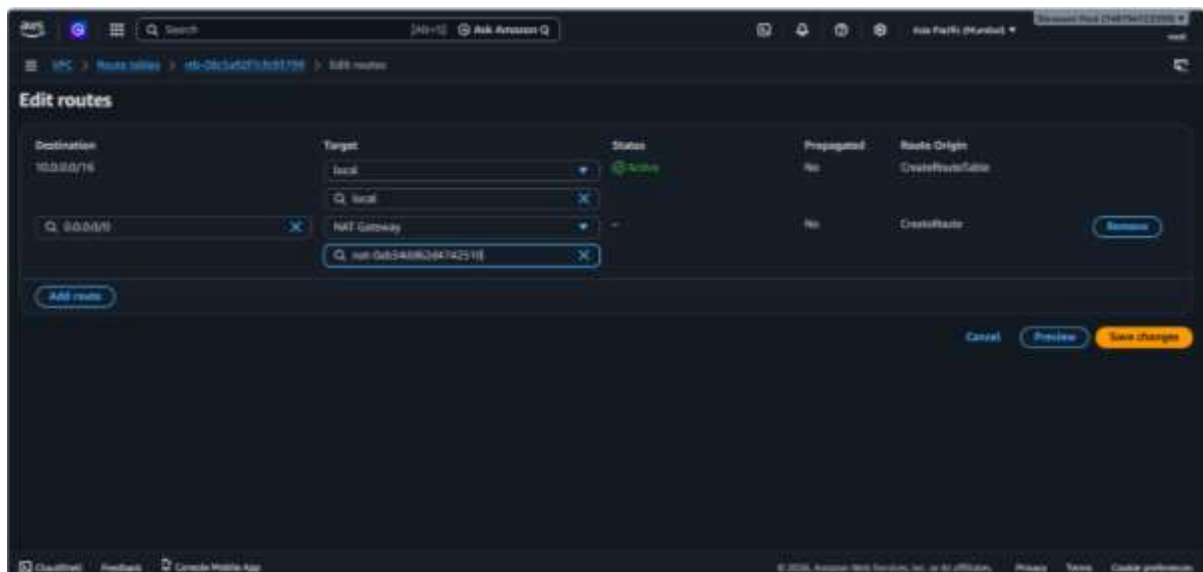


6. Step 5: Route Tables for Private Subnets

Name: my-private-rt-ap-south-1a

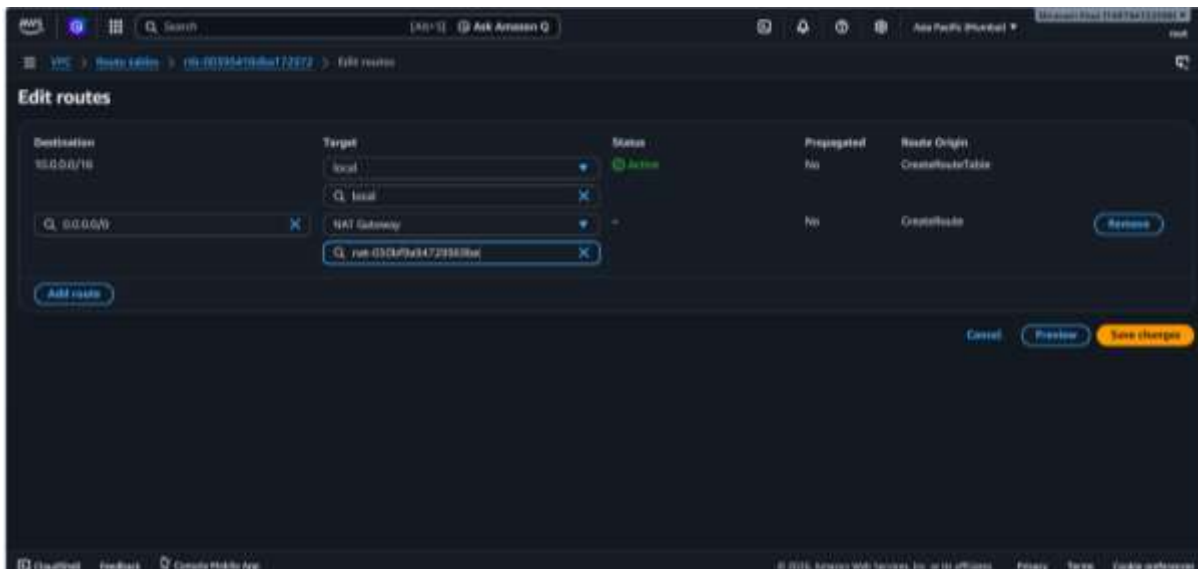
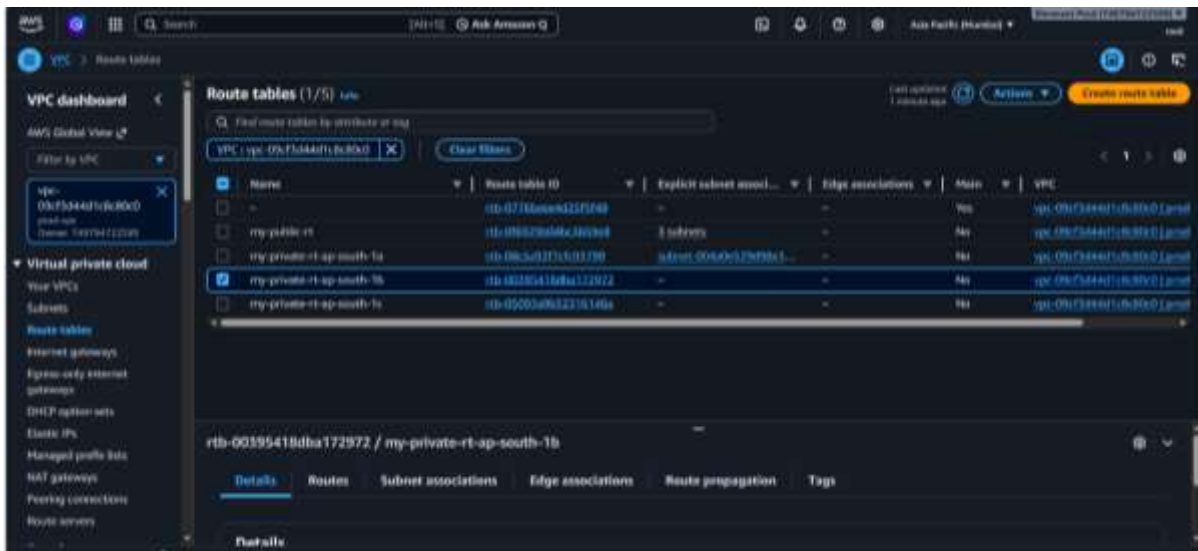
Vpc: prod-vpc





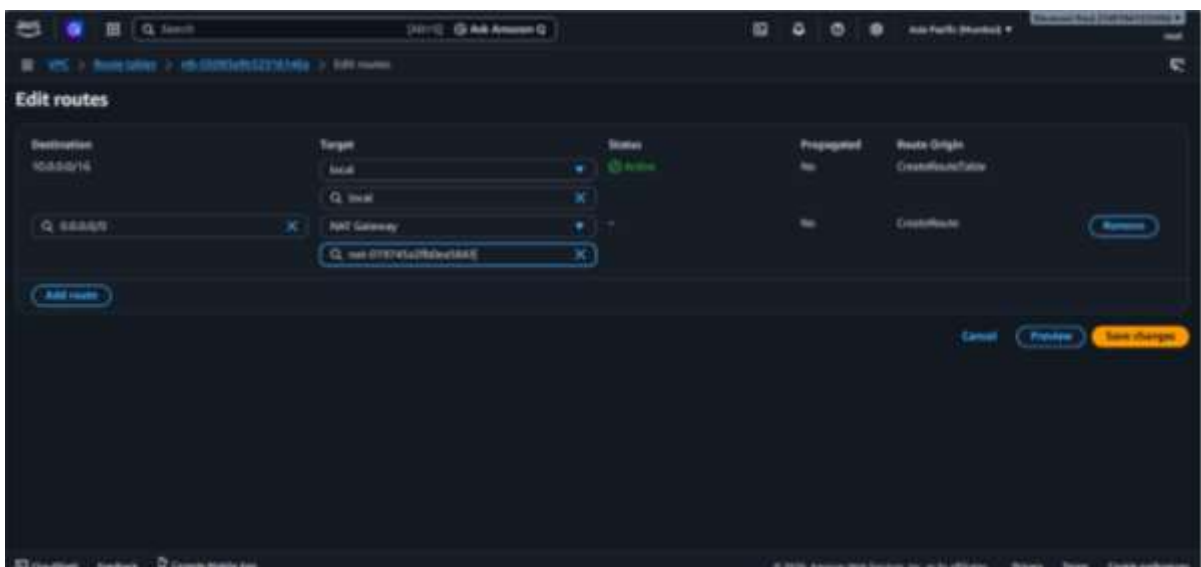
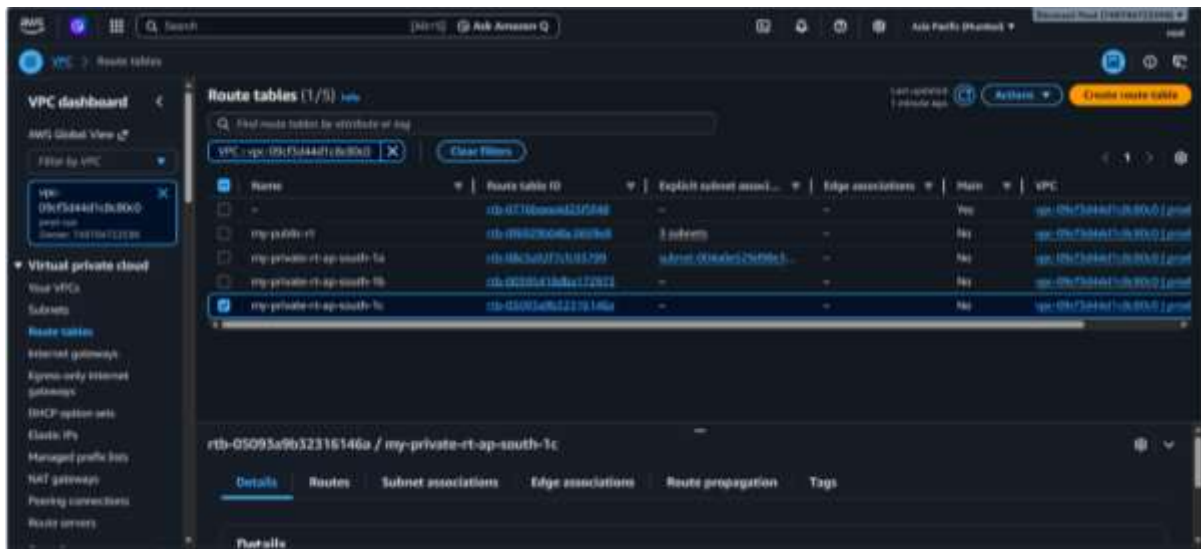
Name: my-private-rt-ap-south-1b

Vpc: prod-vpc



Name: my-private-rt-ap-south-1c

Vpc: prod-vpc



7. Step 6: Final Architecture (VERY IMPORTANT)

AZ-1:

Public-1 → IGW

Private-1 → NAT-1

AZ-2:

Public-2 → IGW

Private-2 → NAT-2

AZ-3:

Public-3 → IGW

Private-3 → NAT-3

8. STEP 7: Verification (DON'T SKIP)

Test 1: Public Subnet:-

- Launch EC2 → should get public IP
- SSH works

Test 2: Private Subnet:-

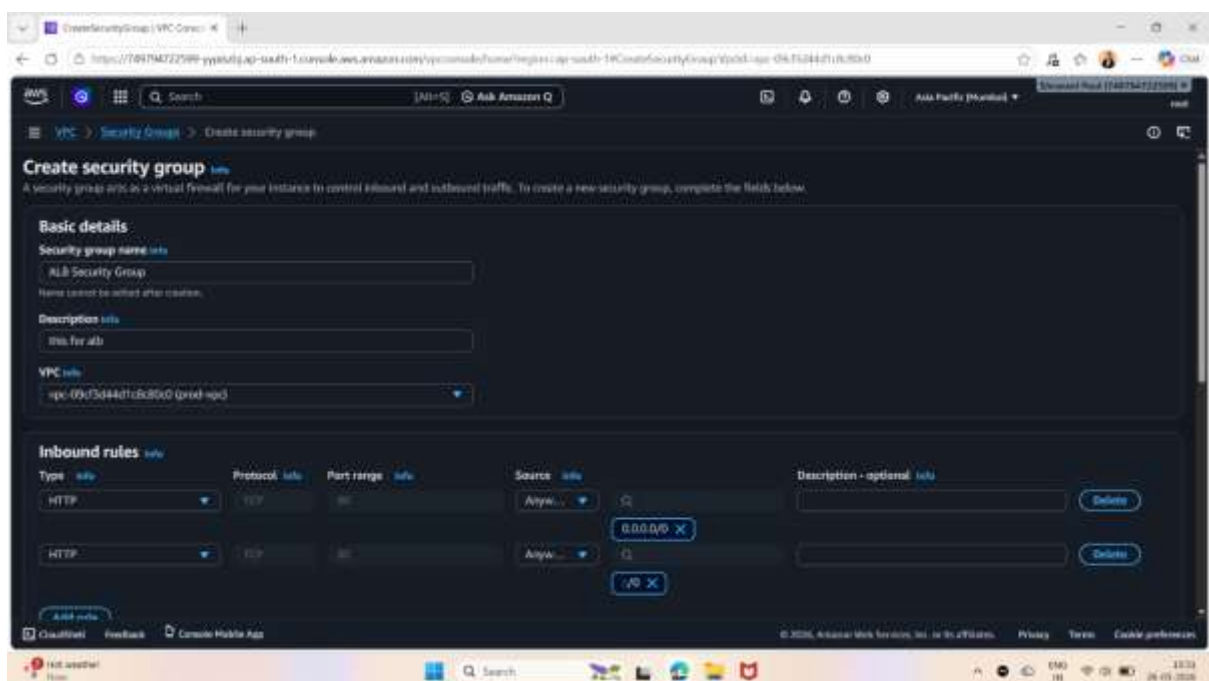
- Launch EC2 → no public IP
- Use NAT → install packages works

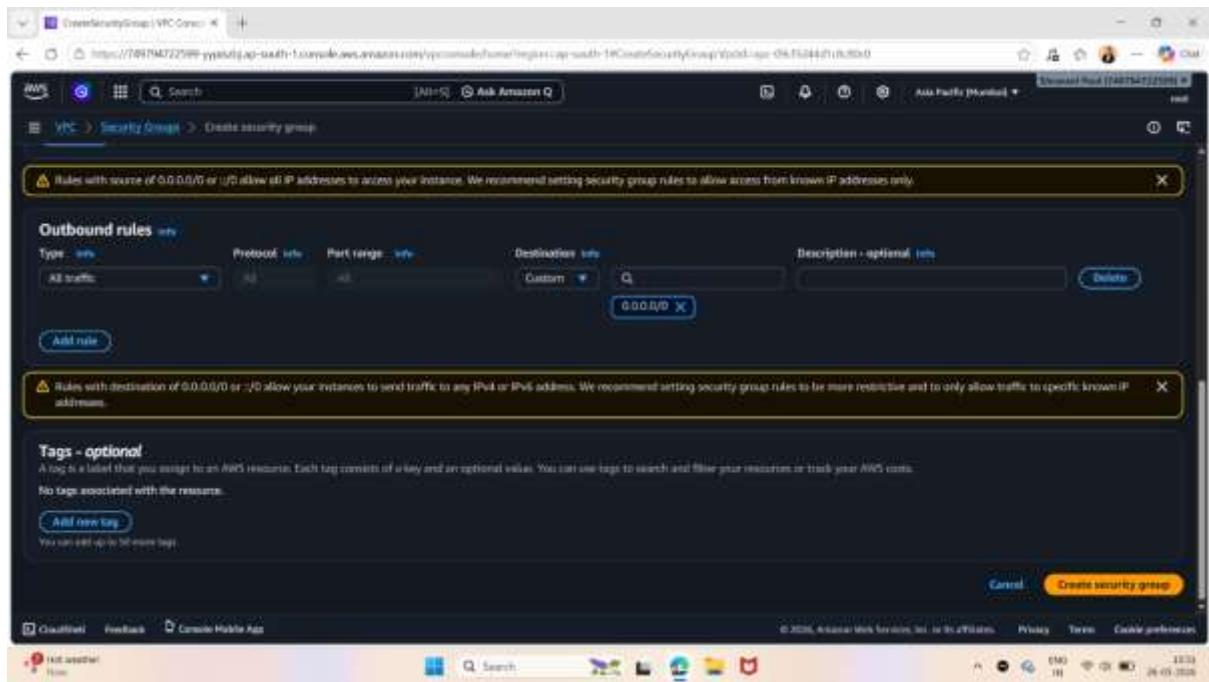
9. Step 8: Security Groups (CRITICAL)

ALB Security Group:-

- Allow:

HTTP (80) → 0.0.0.0/0



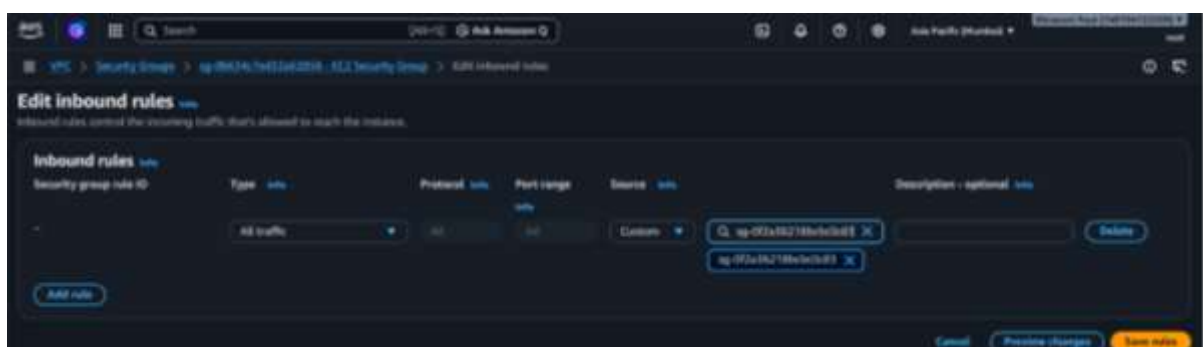


EC2 Security Group:-

- Allow:

HTTP (80) → ALB Security Group

NOT from internet



10.Step 9: Launch Template

Configuration:-

- Name: **web-template**

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared, and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

web-template

Must be unique for this account. Must contain at least one uppercase letter, one lowercase letter, one number, and one special character (for example, @!%, <).>

Template version description

Blank for the web page

Max 255 characters

Auto Scaling guidance [View](#)

Select this if you intend to use the template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

• Template tags

• Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Summary

- Software Image (AMI)
- Virtual server type (Instance type)
- Firewall (security group)
- Storage (EBS volumes)

Cancel Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) [View](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 150,000 of application and OS images

Recently Quick Start

☒ Don't include in launch template ☐ Recently launched

Browse more AMIs

Including AMIs from AWS Marketplace and the Community

Instance type [View](#) [Get pricing](#) [Advanced](#)

Instance type

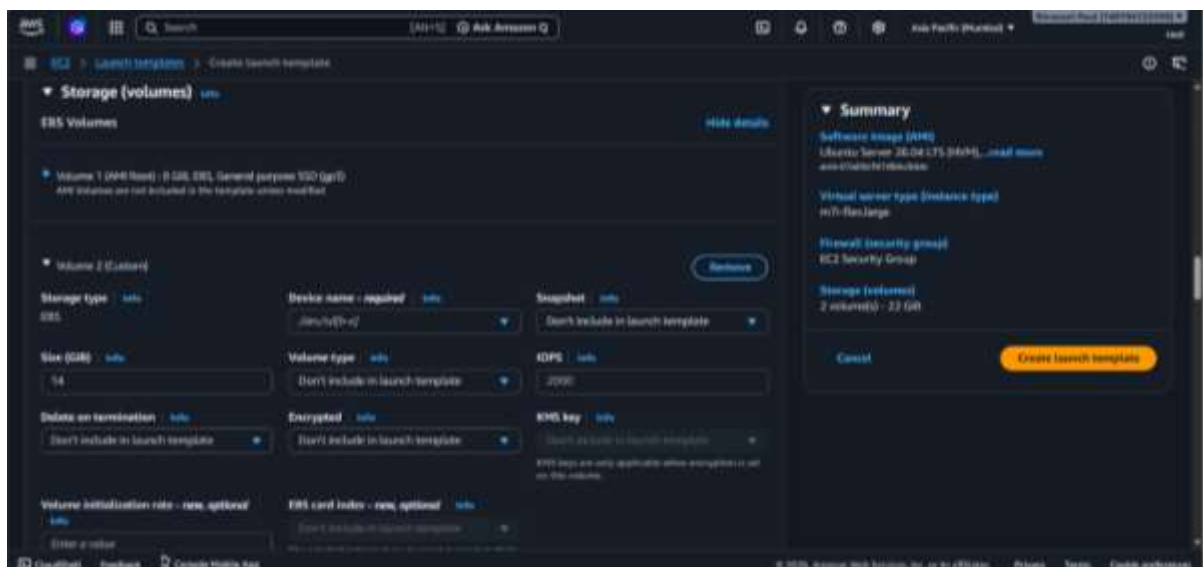
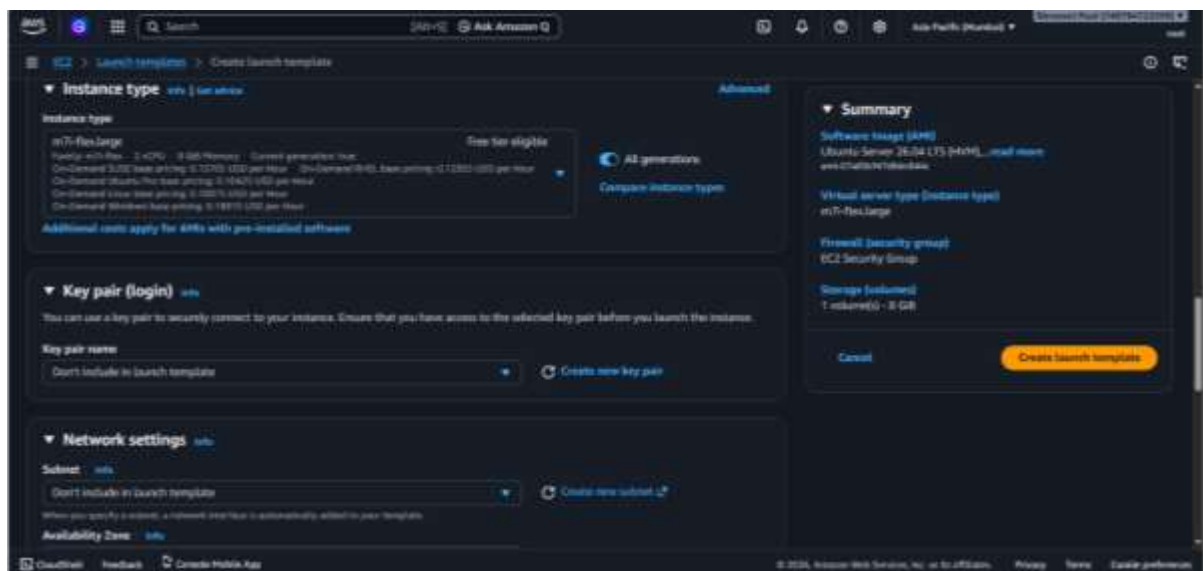
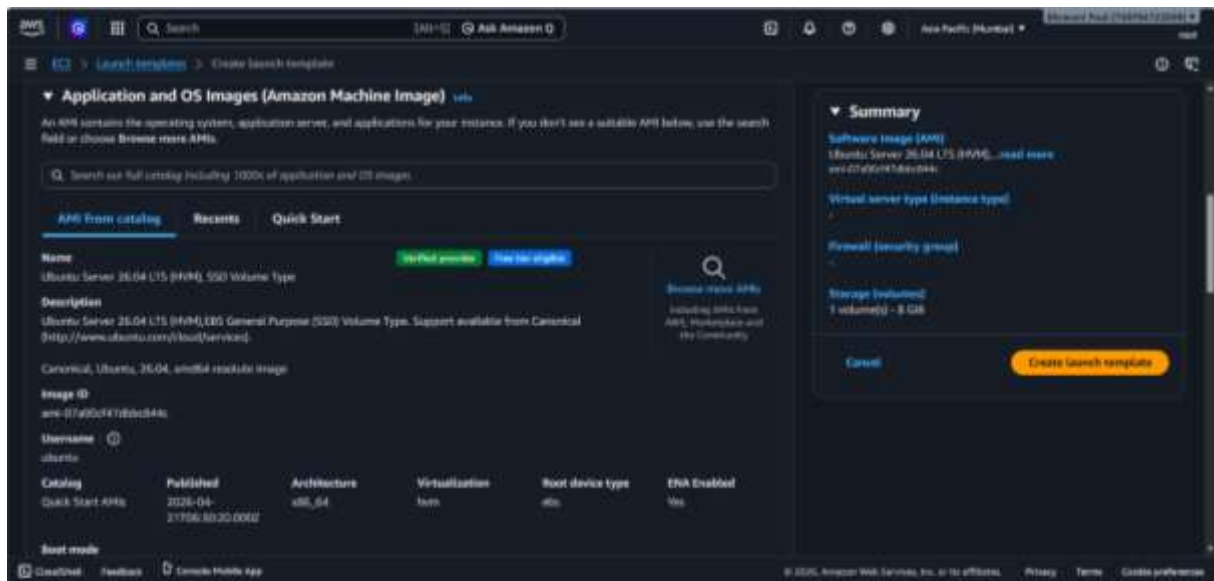
t3.micro

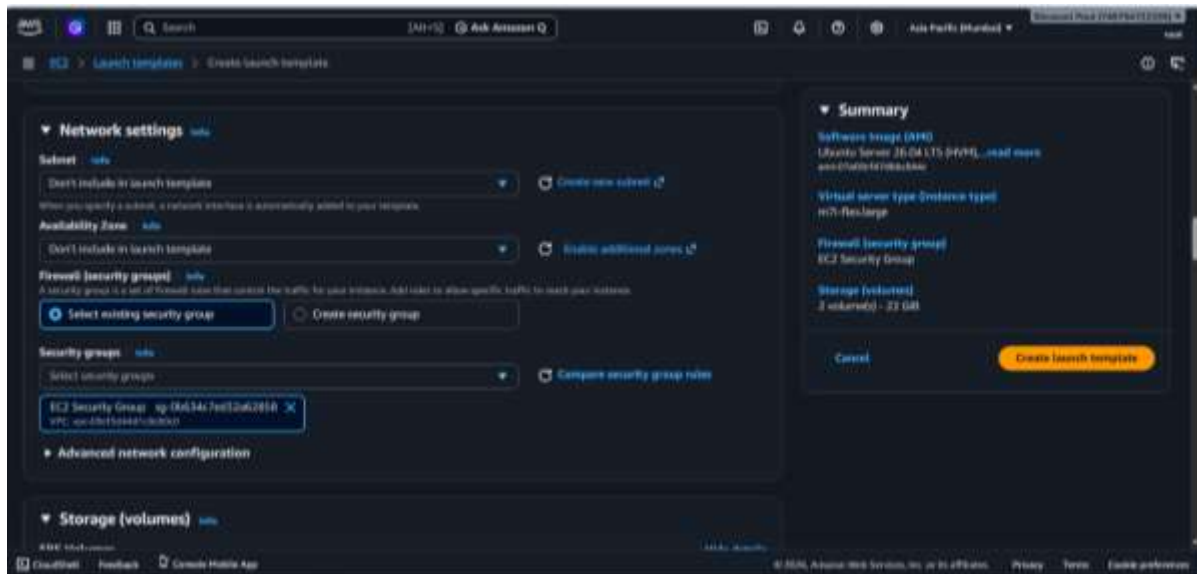
Summary

- Software Image (AMI)
- Virtual server type (Instance type)
- Firewall (security group)
- Storage (EBS volumes)

Cancel Create launch template

- AMI: Ubuntu / Amazon Linux





User Data (Production Demo)

```
#!/bin/bash
```

```
apt update -y
```

```
apt install apache2 -y
```

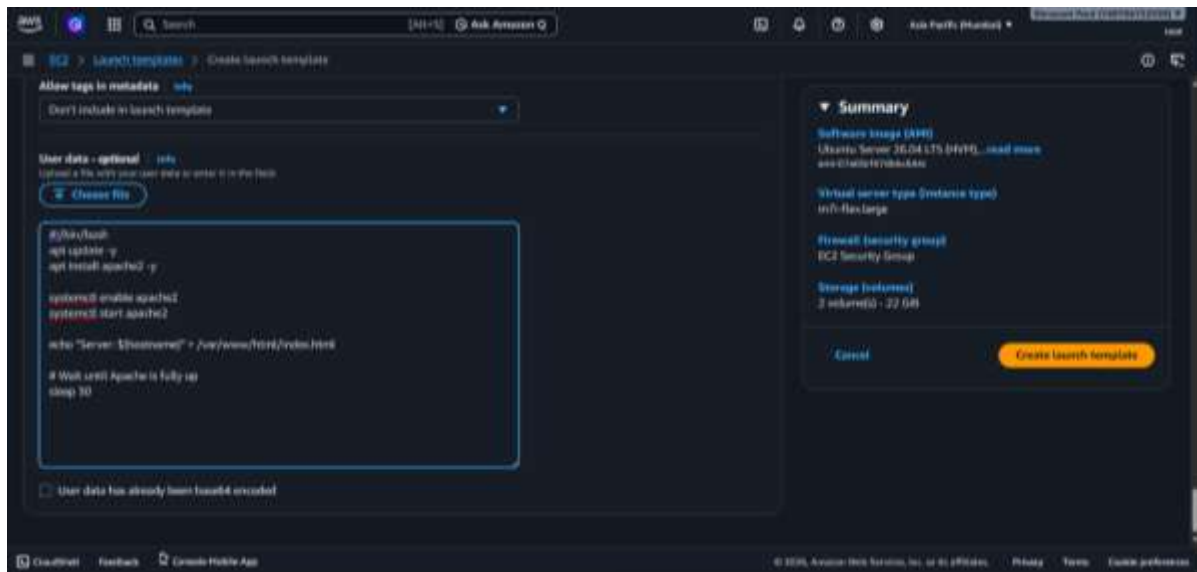
```
systemctl enable apache2
```

```
systemctl start apache2
```

```
echo "Server: $(hostname)" > /var/www/html/index.html
```

```
# Wait until Apache is fully up
```

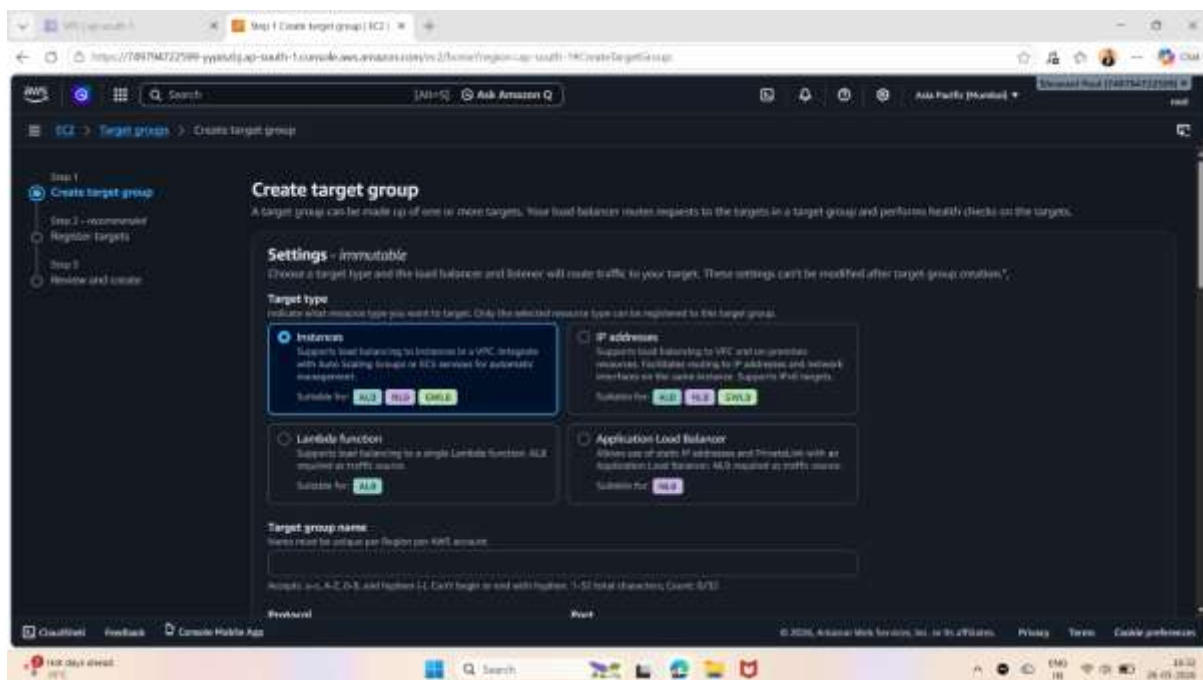
```
sleep 30
```

11. Step 10: Target Group

Configuration:-

- Type: Instance
- Protocol: HTTP
- Port: 80



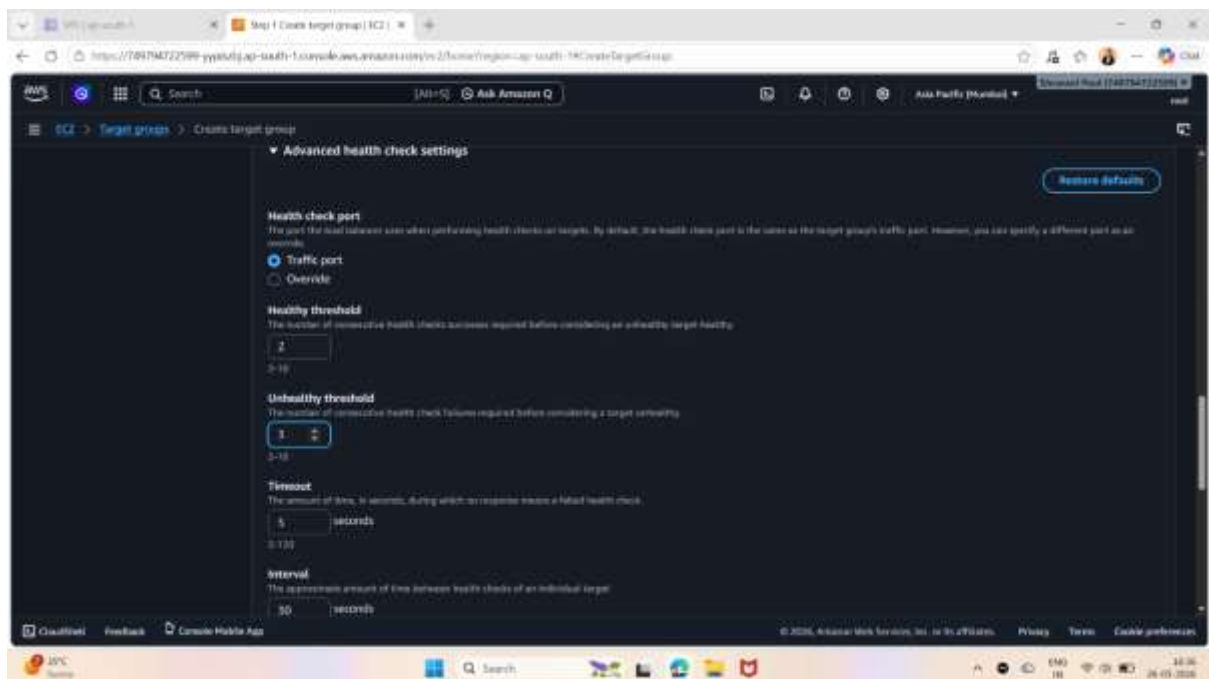
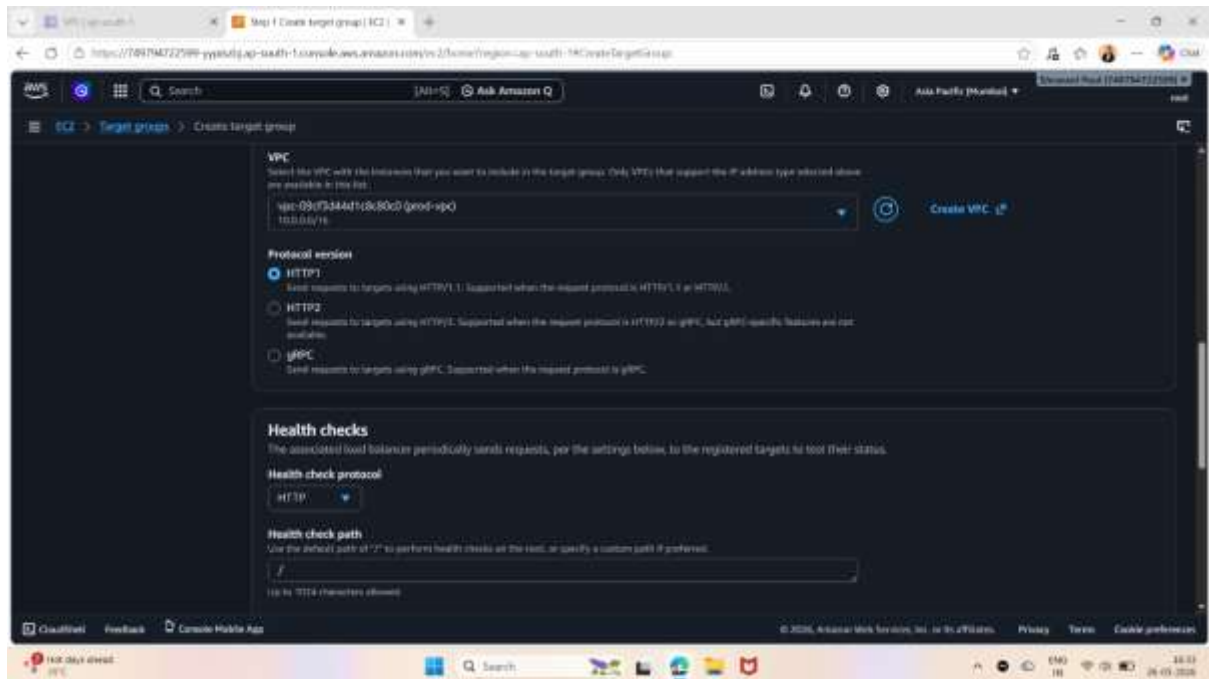
Health Check:-

Path: /

Interval: 30s

Healthy threshold: 2

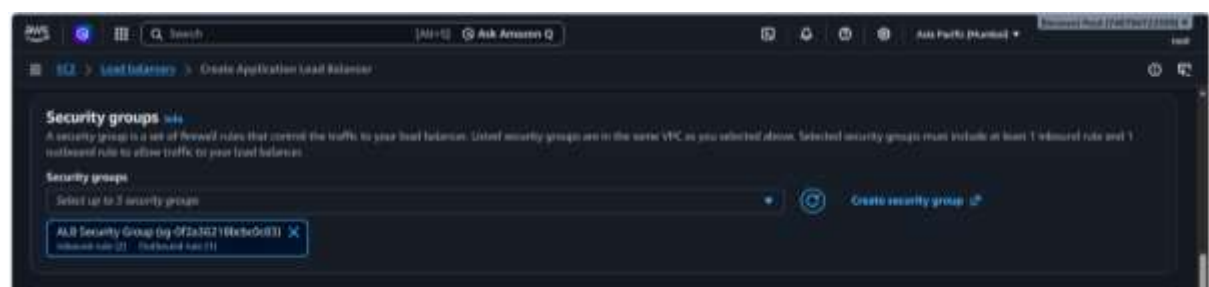
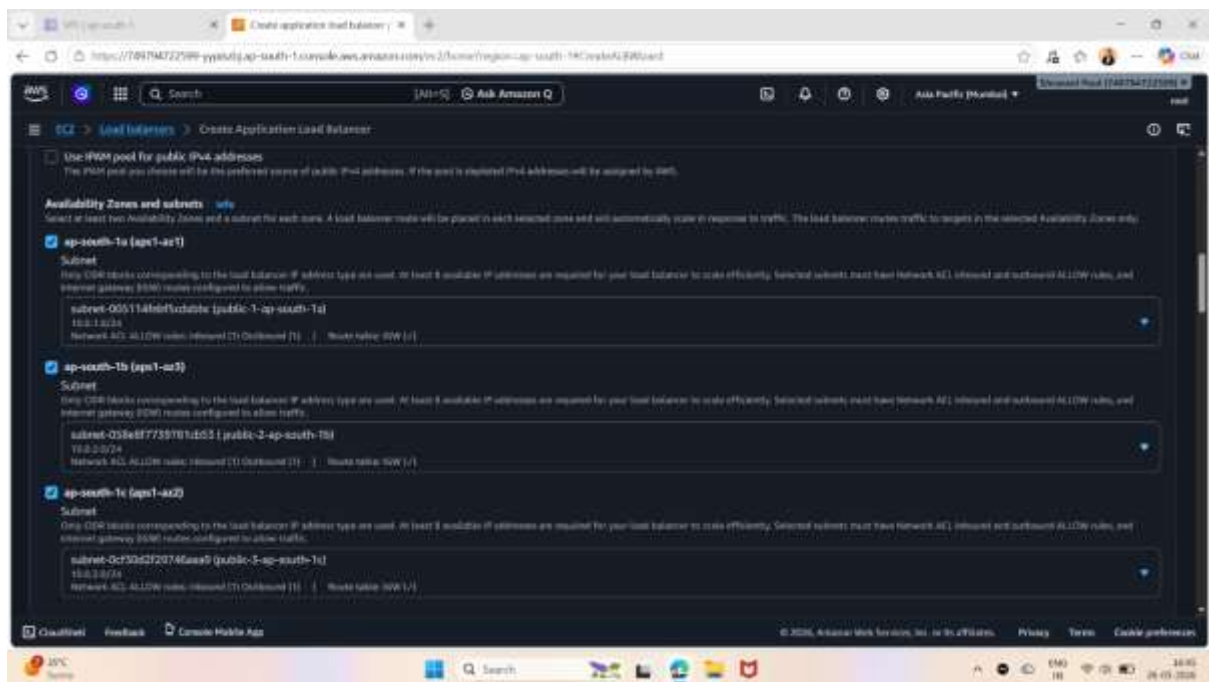
Unhealthy threshold: 3



12. Step 11: Application Load Balancer

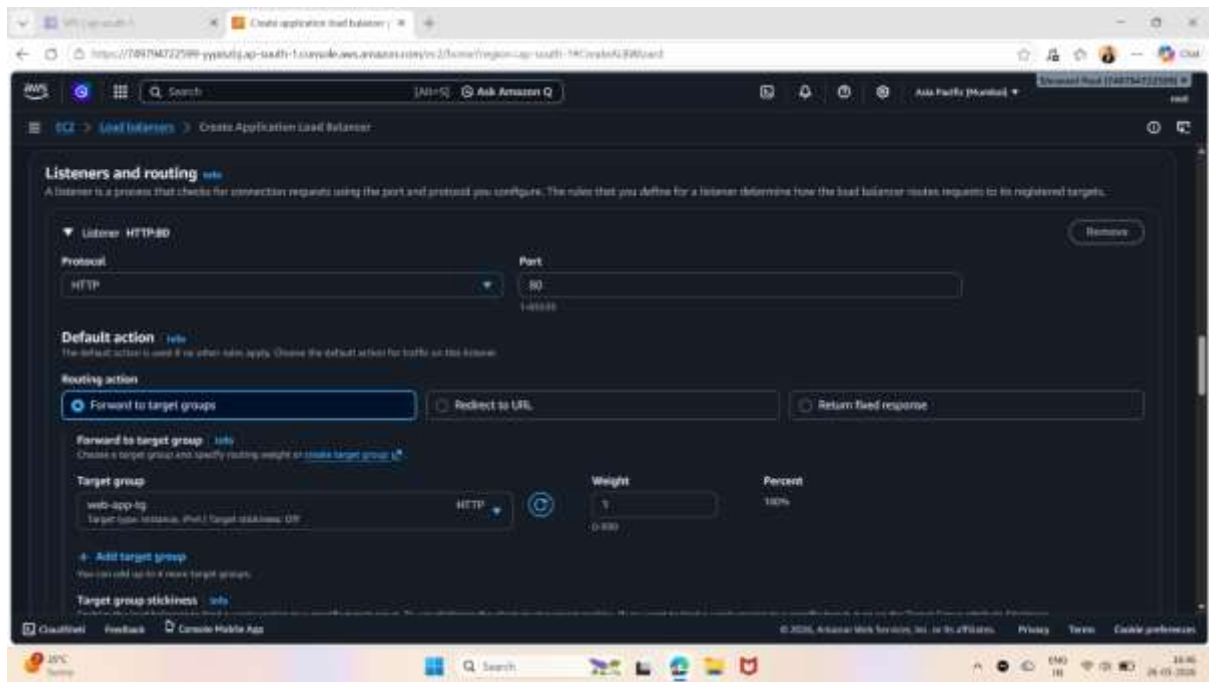
Configuration:-

- Type: Application Load Balancer
- Scheme: Internet-facing
- Subnets: Public subnets
- SG: ALB SG



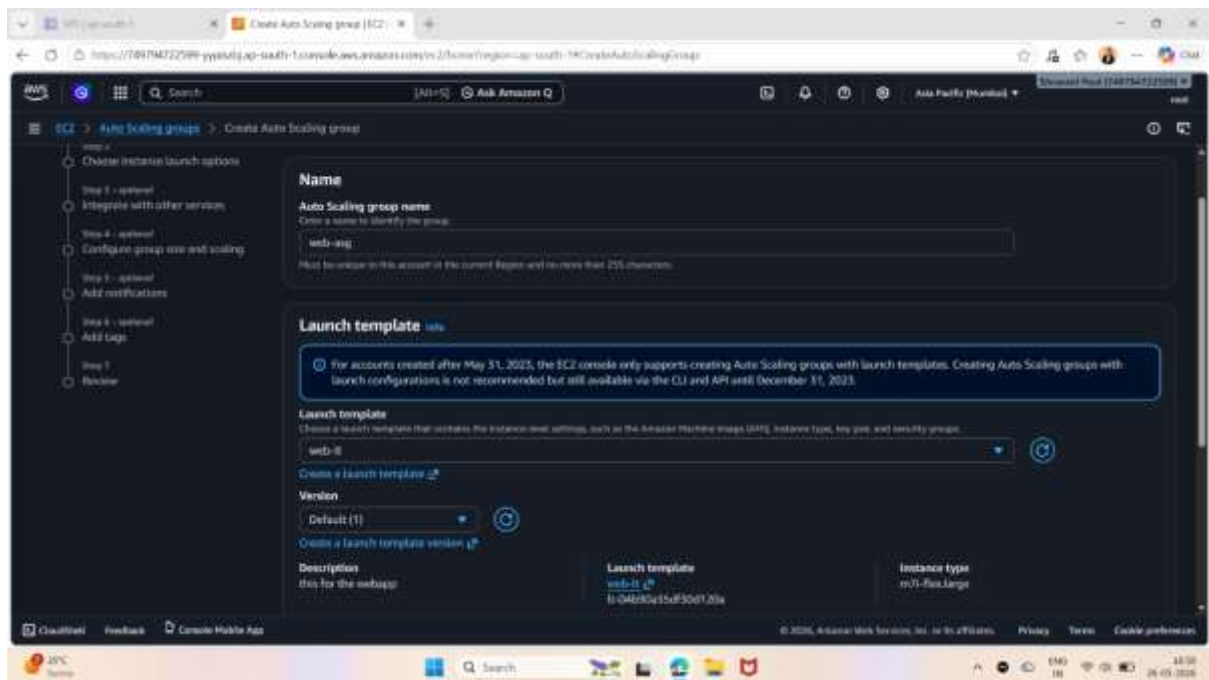
Listener:-

HTTP : 80 → Forward → Target Group

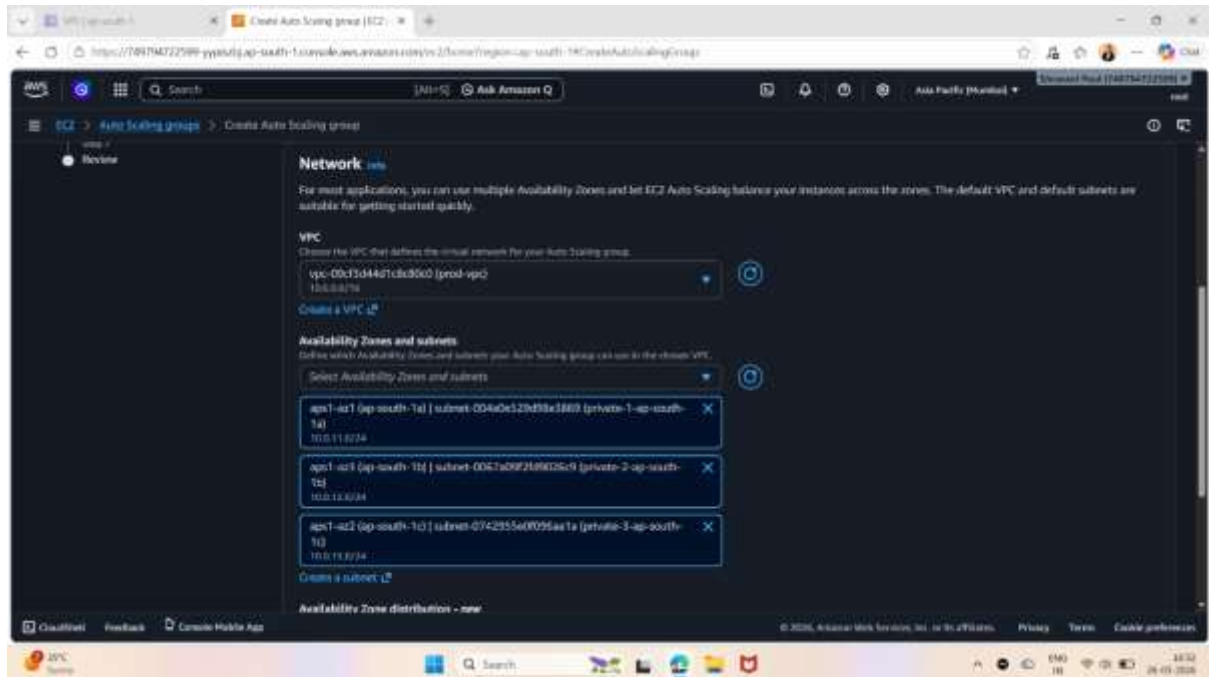


13. Step 12: Auto Scaling Group Configuration:-

- Launch Template: select template

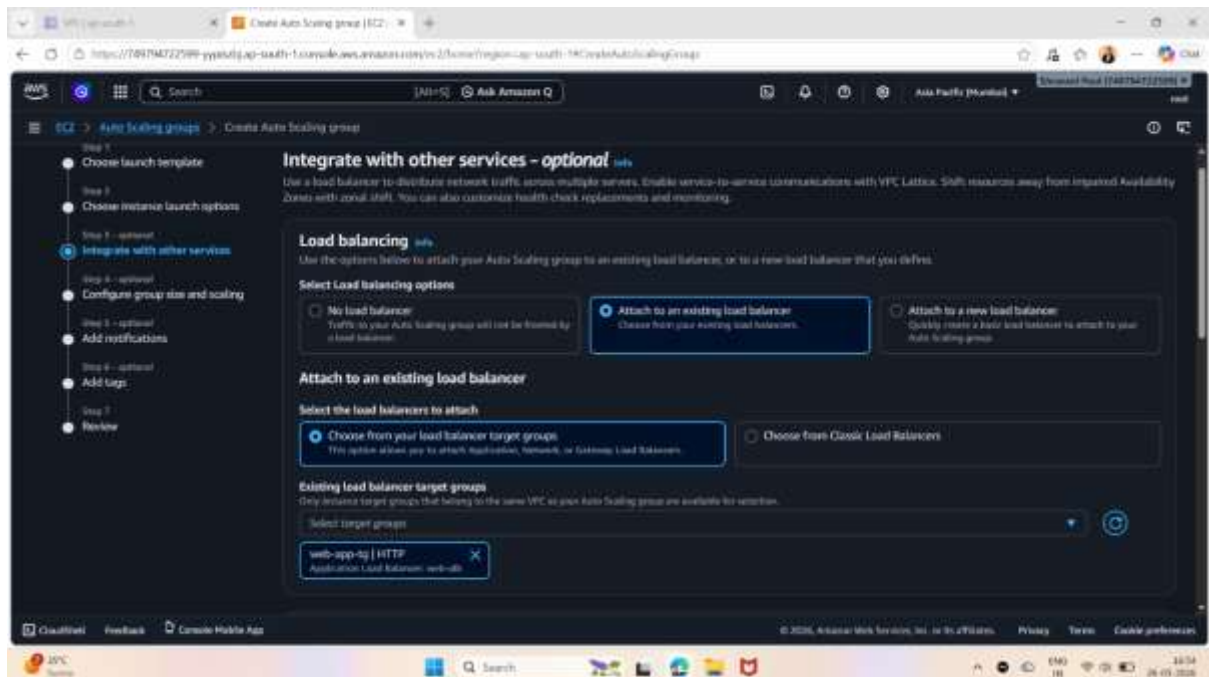


- Subnets: Private subnets



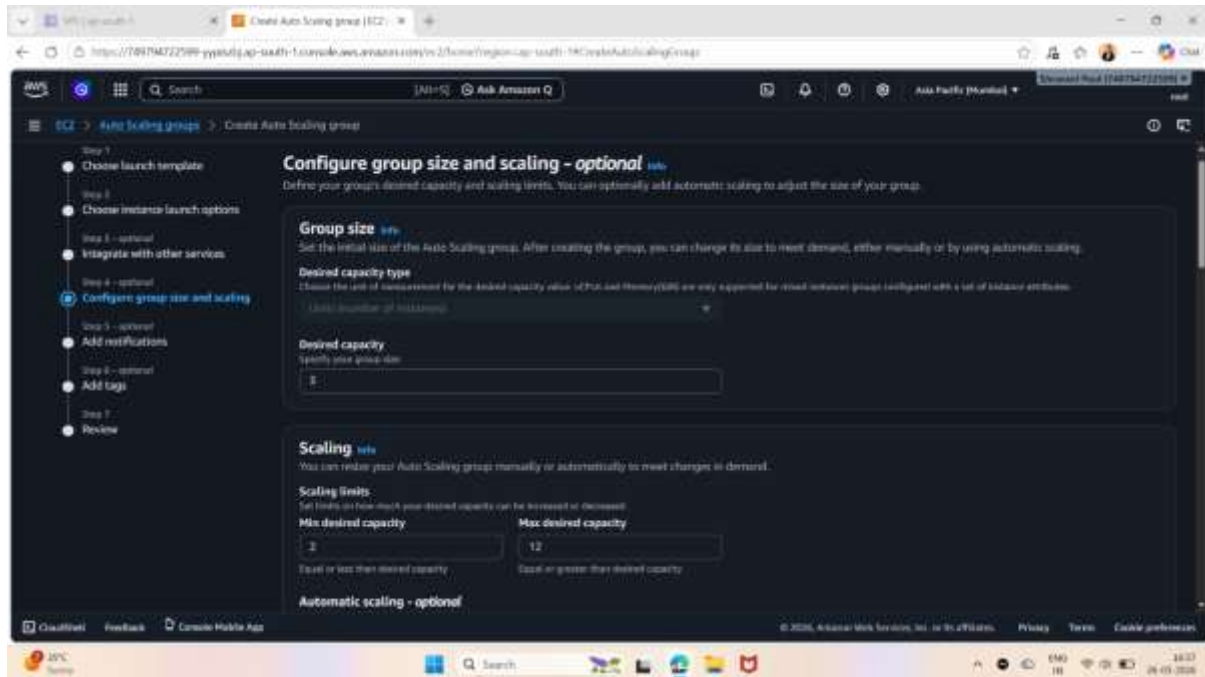
Attach Target Group:-

Mandatory step

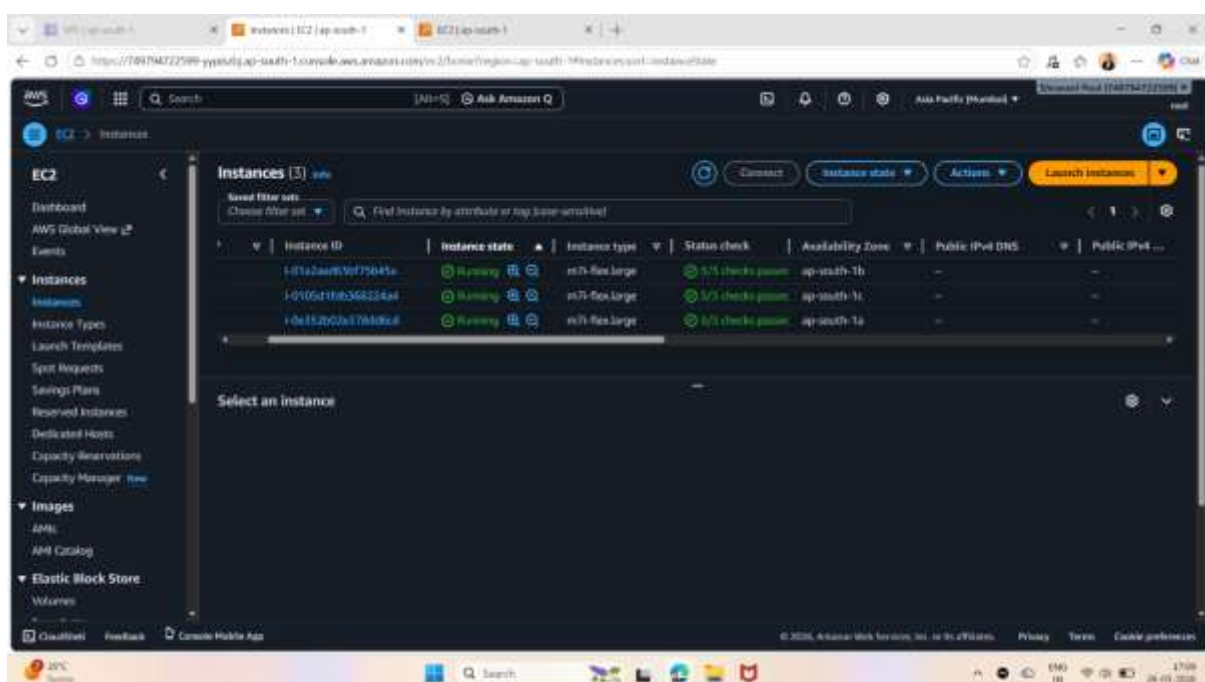


Capacity:-

- Min = 2
- Desired = 2
- Max = 4



After this EC2 Instances are created automatically:



14. Step 13: Testing

Access ALB DNS

